Wrong Prediction, Right Answer: Recovering Evidence from Collapsed LLM Sequence Scores

Anonymous ACL submission

Abstract

A language model can assign the highest likelihood to a wrong answer while its hidden states still encode the correct one. Standard reasoning benchmarks conflate two distinct failure modes—absent answer representations and flawed output scoring—into a single accuracy number. We show these can be disentangled: a two-parameter correction that shifts per-label likelihoods to match an assumed label distribution, fitted from as few as 25 unlabeled examples, recovers 9–34 accuracy points with matching macro-F1 gains across controlled logic, ProofWriter, ANLI, and FOLIO for larger *Qwen3.5* checkpoints ($p < 0.001$ on five of six evaluations). The gains survive on examples a TF-IDF baseline misses and exceed label-count permutation baselines, ruling out lexical shortcuts and label-distribution artifacts. Many wrong answers on logical reasoning tasks are therefore a scoring artifact rather than a reasoning deficit, and benchmark accuracy can systematically underestimate a model's representational capacity.

1 Introduction

Large language models are advancing rapidly in capability and deployment scope (OpenAI et al., 2024; Grattafiori et al., 2024; Anthropic, 2024), and understanding the genuine limits of their reasoning has become a research priority in its own right: knowing *where* and *why* a model fails matters for model comparisons, architectural choices, and training decisions at every scale. Logical reasoning—deductive inference, natural-language entailment, and related three-way decision tasks—has emerged as the primary testbed for this question (Clark et al., 2020; Tafjord et al., 2021; Nie et al., 2020; Han et al., 2024). A wrong final answer is routinely read as direct evidence that a model lacks the underlying reasoning ability. This interpretation motivates two research traditions that look beneath surface accuracy. The first trains lightweight classifiers on a model's internal representations to test what information is encoded before it reaches the output layer (Alain and Bengio, 2018; nostalgebraist, 2020; Belrose et al., 2025), with recent applications to truth directions, factuality, arithmetic, and chain-of-thought answer formation (Bao et al., 2025; Servedio et al., 2025; Maiya et al., 2025; Sun et al., 2025; Yuchi et al., 2026). The second shows that a model's answer choices are not neutral: label frequency, prompt wording, answer phrasing, and training recipe can all push the model toward certain labels regardless of the actual input content (Zhao et al., 2021; Holtzman et al., 2021; Xia et al., 2025; Li et al., 2025a; Sanz-Guerrero and von der Wense, 2025; Li et al., 2025b; Wang and Liu, 2025; Huang et al., 2026; Xiao et al., 2025; Nakkiran et al., 2025).

Each tradition reveals part of the picture, but each also stops short of answering the diagnostic question: when a model gives a wrong answer, is the correct answer absent from its representations, or is a correctable output-scoring bias preventing it from surfacing?

Probing shows that information *could* be decoded from hidden states by an auxiliary classifier, but probe success does not imply that the frozen model can recover the same information through its own output scoring (Hewitt and Liang, 2019; Voita and Titov, 2020; Pimentel et al., 2020; Belinkov, 2022). Indeed, models are known to encode correct answers in their internal representations while outputting the wrong token (Burns et al., 2023)—a gap that probing can detect but not close. Calibration faces a complementary limitation: it can raise overall accuracy by shifting the label distribution, yet a corrected label histogram is not the same as correct per-example decisions—the right aggregate counts can arise from fixing the wrong individual examples.

To close this blind spot, we introduce a per-example comparison of scoring methods (Figure 1)

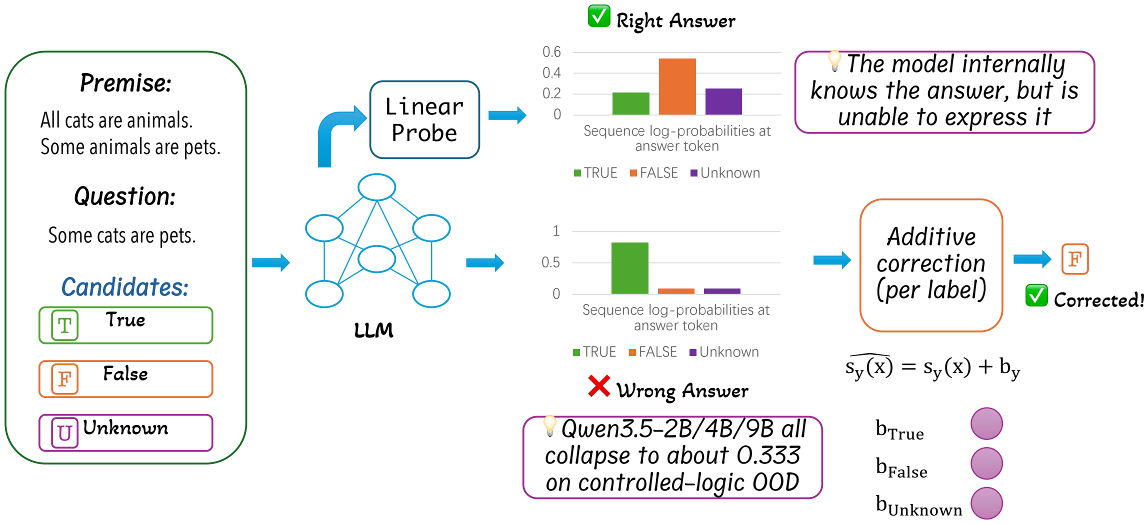

Figure 1: **Correcting sequence score collapse in logical reasoning.** While raw sequence log-probabilities frequently yield the wrong answer due to score collapse, a linear probe demonstrates that the correct semantic representation exists internally. Applying an unsupervised additive correction to the candidate labels ($s_y(x) = s_y(x) + b_y$) shifts the output scores to bypass this bottleneck and express the corrected answer.

that places internal-representation classifiers and the model's own answer scoring side by side on the same examples. We apply a minimal bias correction—two additive per-label offsets fitted without any target labels—and accept a gain as evidence of preserved answer information only if it survives controls for lexical shortcuts, overall label-distribution matching, and answer-surface variation.

Across controlled logic, ProofWriter, ANLI, and FOLIO, larger *Qwen3.5* checkpoints (Qwen Team, 2025) show a consistent and large gap between internal-representation accuracy and what the model chooses when scoring complete answer strings. Fitting per-label offsets from as few as 25 unlabeled examples—without any target labels—recovers 9–34 accuracy points and matching macro-F1 on held-out examples, including examples that a simple bag-of-words classifier misses. Counterexample models and high-accuracy baselines delimit where the same analysis does not apply. Many wrong answers on logical reasoning benchmarks therefore reflect a correctable output-scoring failure, not the absence of reasoning ability.

Our contributions are:

- We identify a diagnostic conflation in LLM reasoning evaluation: benchmark accuracy merges absent answer knowledge with output-scoring failures into a single number, masking what a model actually represents.
- We introduce a per-example scoring comparison that jointly tests hidden-state decodability, vocabulary-head output, and complete-sequence likelihood, with a two-parameter unsupervised label-bias correction and a full suite of controls to scope the recovery claim.
- We demonstrate that larger *Qwen3.5* checkpoints—including instruction-tuned variants—preserve per-example answer evidence in their output scores even when raw likelihood collapses, recovering 9–34 accuracy points across four reasoning benchmarks; scope controls identify where this conclusion does not hold.

2 Related Work

Probing versus output scoring. Linear probes trained on frozen hidden states have established that models encode substantial task-relevant information before the final output token (Alain and Bengio, 2018; nostalgebraist, 2020; Belrose et al., 2025), with extensions probing for truth directions, factuality, arithmetic steps, and chain-of-thought answer formation in large LLMs (Bao et al., 2025; Servedio et al., 2025; Maiya et al., 2025; Sun et al., 2025; Yuchi et al., 2026; Kudo et al., 2026). These studies share a methodological gap that limits their diagnostic reach: probe accuracy is measured against ground truth, but the model's own candidate-answer scoring is never placed beside it on the same examples. This omission is consequential. Control-task and MDL analyses show that high probe accuracy can arise from statistical features in the representations that the model's own prediction path never uses (Hewitt and Liang, 2019; Voita and Titov, 2020; Pimentel et al., 2020;

Belinkov, 2022); probe success therefore does not imply that the frozen model's output scoring accesses the same information. Burns et al. (Burns et al., 2023) provide the sharpest evidence that the gap is real: models can internally represent the correct answer while outputting the wrong token. We take this as our starting point: if the gap between internal representation and output scoring is real, the natural next question is what prevents the correct answer from reaching the output.

Output bias and per-example recovery. LM predictions over fixed answer choices are systematically distorted by label frequency (Zhao et al., 2021), surface-form competition—where longer answer strings receive lower probability simply because they span more tokens (Holtzman et al., 2021)—and prompt wording or instruction tuning (Xia et al., 2025; Li et al., 2025a; Huang et al., 2026). A range of post-hoc corrections address these biases: verbal normalization, diagonal scaling (Wang and Liu, 2025; Sanz-Guerrero and von der Wense, 2025), calibration-aware fine-tuning (Xiao et al., 2025), and judge-bias mitigation (Li et al., 2025b; Nakkiran et al., 2025). A shared limitation runs through this work: calibration gains are evaluated by aggregate accuracy or expected calibration error, neither of which distinguishes a correction that fixes the right examples from one that merely reshapes the label distribution. This per-example blind spot is compounded by the softmax bottleneck (Yang et al., 2017): a frozen output layer need not expose all task-relevant directions in the hidden state, so label-frequency artifacts can dominate the prediction even when the model internally encodes the correct answer. We address this directly with per-example gain counts and label-count permutation baselines that verify corrected decisions fix the right instances.

The two gaps above—probing without output comparison, and calibration without per-example verification—are most visible on benchmarks whose design properties allow each pipeline component to be audited independently.

Logical-reasoning benchmarks as a diagnostic surface. Controlled deduction tasks (Clark et al., 2020; Saparov and He, 2023) and natural-language benchmarks—ProofWriter (Tafjord et al., 2021), FOLIO (Han et al., 2024), ANLI (Nie et al., 2020)—are routinely used to compare models and support claims about the effects of scale and post-training. Their design properties—known ground truth, balanced label distributions, controllable proof depth—make them ideal for auditing individual components of a prediction pipeline, yet prior work uses them only as leaderboards: accuracy is the terminal metric, and a wrong answer is read as direct evidence of absent reasoning. This interpretation conflates two failure modes and makes comparative claims unreliable: a model that encodes the correct answer but fails at output scoring is indistinguishable in accuracy from one that never represents the answer at all. We repurpose these benchmarks as a controlled diagnostic surface to pull the two modes apart—a repurposing that prior evaluations have not pursued.

Measurement	Question tested	Supporting observation
Hidden-state probe	Is the answer linearly decodable at the answer position?	Probe accuracy exceeds complete-sequence likelihood on the same examples.
Vocabulary-head predictions	Does the frozen output layer expose the same information as the probe?	Probe accuracy remains above vocabulary-head and first-token accuracy.
Complete-sequence likelihood	Does the model's own answer-string score select the right label?	Likelihood collapses to one label while probes remain informative.
Label-bias correction	Do relative sequence scores preserve per-example evidence after label offsets are removed?	Per-example accuracy and macro-F1 improve without any target labels.
TF-IDF and permutation controls	Are gains due to lexical shortcuts or only label-count matching?	Gains persist on TF-IDF-missed examples and exceed label-count permutation baselines.

Table 1: **Measurement logic for the scoring comparison.** Each row corresponds to one link in the evidence chain: hidden-state decodability, vocabulary-head predictions, complete-sequence scoring, unsupervised label-bias correction, and shortcut or label-count controls.

3 Method

Figure 1 and Table 1 give the overview; this section defines each component. To localize *where* a final-answer error arises, the method builds a three-step diagnostic chain: (i) establish whether the answer is encoded in hidden states; (ii) test whether the model's own output scoring exposes that information; (iii) apply a minimal unsupervised correction and verify that any recovery holds at the per-example level. No training objective is added; every component operates on frozen model outputs.

Task and complete-sequence likelihood. To establish the scoring signal that all comparisons build on, we score each candidate answer as a full continuation under the frozen model. Each example has a prompt x and a label $y \in \{\texttt{true}, \texttt{false}, \texttt{unknown}\}$ (or the analogous ANLI triple); for each label we fix a candidate string a_y after an answer prefix r and compute

$$s_y(x) = \log p_\theta(a_y \mid x, r).$$

No task head or classifier is added; the frozen LM assigns each label a score, and the raw prediction is $\arg\max_y s_y(x)$. Length normalization (dividing by the number of answer tokens) is applied when candidate-string lengths vary and retained as a robustness check.

Hidden-state probe and vocabulary-head predictions. To separate hidden-state evidence from an output-scoring failure, we cache the hidden state $h^a(x)$ at an explicit answer position—the token just before the model begins emitting `true`, `false`, or `unknown`. A supervised logistic-regression probe tests whether the gold label is linearly decodable from $h^a(x)$. At the same position, we also apply the frozen unembedding matrix W_U to produce vocabulary-head predictions. This position-matched design separates what is linearly decodable in the model's internal states from what the model's own scoring expresses: a large gap between probe and vocabulary-head accuracy identifies answer information that output scoring fails to surface.

Label-bias correction. We model the raw score as $s_y(x) = \tilde{s}_y(x) + b_y$, where $\tilde{s}_y(x)$ is the task-relevant component and b_y is a per-label additive artifact (e.g., label-frequency bias accumulated during pre-training or instruction tuning). The raw prediction $\arg\max_y s_y(x)$ is distorted by differing b_y values; subtracting them restores the underlying ranking. For a target domain with an unlabeled in-domain score set D_{id}, we estimate b_y by fitting additive offsets on top of the frozen scores,

$$\hat{y}_b(x) = \arg\max_y \; s_y(x) + b_y,$$

giving two free parameters for three-way tasks after removing a common offset. Let π be the assumed target label distribution and $q_b(y; D_{\text{id}})$ the induced prediction fraction on the fit set. Histogram matching provides a natural estimator: since the marginal prediction distribution under the true task-relevant scores should reflect the label prior π, deviations from π identify the per-label biases to remove,

$$b^\star = \arg\min_b \sum_y \left(q_b(y; D_{\text{id}}) - \pi_y\right)^2.$$

The fit sees only candidate-answer scores—no gold labels—and the offsets are frozen before held-out evaluation. The correction is a class-conditional additive shift, strictly less expressive than temperature scaling: it adjusts only the decision thresholds between labels and cannot alter the relative ordering of scores within a label. This low capacity is by design—it ensures that any held-out gain reflects a correctable label-level artifact rather than a hidden task learner. When the target distribution is unknown, varying π serves as a sensitivity check.

Evaluation protocol. To ensure the correction never sees target labels before evaluation, each model–domain pair follows a fixed pipeline: (i) fix answer strings, (ii) compute frozen sequence scores, (iii) fit per-label offsets on unlabeled in-domain scores, (iv) freeze offsets, and (v) compare raw and corrected decisions on a disjoint held-out set. A result counts as evidence only when it has positive per-example gains *and* survives the controls below. Grid search, bootstrap, and probe details are in Appendix B.

Controls. To rule out two direct alternative explanations—lexical shortcuts and label-count matching—three controls accompany every main result. The *per-example gain* counts examples the corrected rule fixes minus those it breaks; a positive value confirms real per-example improvement, not redistribution across labels. *TF-IDF-missed examples* restrict evaluation to examples a bag-of-words classifier fails, blocking lexical recovery. *Label-count permutation baselines* randomly permute corrected predictions; exceeding this baseline rules out label-distribution-only gains. We report accuracy, macro-F1, per-example gains, and permutation gaps throughout; bootstrap intervals and deterministic subsampling check stability.

Evidence selection. The main text reports experiments that answer one of three questions: (i) does sequence scoring fail despite probe evidence at the answer position, (ii) does an unsupervised label-bias correction recover per-example accuracy, and (iii) do those corrected decisions survive shortcut and label-count controls? Component ablations, sweeps, and prompt grids are in the appendix; the experiments follow the same order as the claim so that no result is read as evidence for the next without the intervening check.

4 Experiments

4.1 Setup

Three design choices follow from the diagnostic goal. Benchmarks are selected to make each pipeline component separately auditable. The model family is held fixed so that scoring-method variation is not confounded with model choice. The correction is always fit on data disjoint from the evaluation set, so any recovery is unsupervised with respect to target labels and any held-out gain reflects a genuine per-example effect.

Datasets. To audit each pipeline component independently, we need benchmarks where label balance, lexical variation, and proof depth are separately controllable. Our controlled-logic dataset satisfies this: a balanced three-way deduction task with nonce terms, shuffled facts and rules, and positive/negative distractors, yielding depth, lexical, stress, and counterfactual splits from a 1,000-example in-domain base. External benchmarks test generalization: ProofWriter OOD (Tafjord et al., 2021) and ANLI R2 (Nie et al., 2020) cover natural-language rule following and adversarial NLI; FOLIO (Han et al., 2024) serves as a first-order logic transfer check (fit on 203 validation examples, evaluated on 1,001 held-out). All non-FOLIO held-out evaluations use $\geq$1,000 examples.

Models. To isolate scoring-method effects from model-choice variation, we trace one family—*Qwen3.5*—through every comparison. Base checkpoints (Qwen Team, 2025) test whether the probe-vs.-sequence gap appears before instruction tuning; post-trained *Qwen3.5-2B/4B/9B* test whether the same failure persists after it. Model choices are diagnostic, not leaderboard-driven. Additional controls (*Pythia-2.8B/12B*) and cross-task ensemble experiments are in the appendix.

Train/test separation. The offset-fit set is always disjoint from the evaluation set: controlled logic and ProofWriter fit on in-domain data and evaluate on shifted or OOD splits; ANLI fits on R1 and evaluates on R2; FOLIO fits on 203 validation examples and evaluates on 1,001 held-out. Sample-efficiency experiments refit on smaller subsamples and evaluate on the full held-out set.

The central result: for larger *Qwen3.5* checkpoints, two per-label offsets fitted from 25 unlabeled examples recover 9–34 accuracy points on three-way reasoning benchmarks, with per-example gains that survive TF-IDF, label-count, prior, and verbalizer controls. We build to this result in a cumulative chain—probe gap (§4.2), bias correction (§4.3), sample efficiency (§4.4), controls (§4.5)—so that each claim is supported before the next is introduced; a result that fails any link is a scope control.

4.2 Readout gap: sequence scoring fails while answer-position evidence remains

Hypothesis. If candidate-answer likelihood is the failing step, answer evidence should still be present in hidden states even when sequence scoring collapses (i.e., concentrates all predictions on one label). Specifically, hidden-state probes should substantially outperform sequence-scoring accuracy on the same examples. Table 2 confirms this for post-trained *Qwen3.5-2B/4B/9B* (base-model results, which show the same pattern, are in Appendix D). Post-trained *Qwen3.5-2B* reaches probe accuracies of 0.918 (in-domain), 0.583 (depth-shifted), and 0.661 (lexical OOD), while complete-sequence likelihood stays at chance (0.333) on all three splits. *Qwen3.5-4B/9B* preserve the same gap: probes exceed 0.80 on most splits while sequence likelihood remains near chance. Answer absence is therefore not the only explanation for wrong sequence predictions—the hidden state at the answer position can hold decodable evidence while the model's own scoring stays collapsed.

The probe establishes that the answer is linearly decodable from the hidden state; the next question is whether the model's own sequence scores still carry useful per-example evidence once label-frequency bias is removed.

4.3 Label-bias correction recovers per-example evidence

Hypothesis. If sequence scoring fails mainly because of per-label frequency bias, then a two-parameter unsupervised correction should recover held-out decisions *at the example level*: macro-F1 should move with accuracy and per-example gains should be positive—not merely the overall label distribution. Table 3 and Figure 2 confirm this. Fitting only additive per-label offsets from unlabeled in-domain scores raises *Qwen3.5-4B/9B* on the synthetic lexical split from chance to 0.570/0.602 and on ProofWriter to 0.653/0.678; all six *Qwen3.5* rows reach significance at $p \leq 0.035$ (five at $p < 0.001$). ANLI is harder but still positive for *Qwen3.5-4B*, moving from 0.478 to 0.571

Model	Split	Prompt Probe	Answer Probe	Slot W_U	Vocab Head	Seq.
Qwen3.5-2B	`id`	0.908	0.918	0.333	0.333	0.333
Qwen3.5-2B	`depth`	0.580	0.583	0.333	0.333	0.333
Qwen3.5-2B	`lexical`	0.516	0.661	0.333	0.333	0.333
Qwen3.5-4B	`id`	0.951	0.340	0.580	0.572	0.333
Qwen3.5-4B	`depth`	0.761	0.336	0.521	0.515	0.333
Qwen3.5-4B	`lexical`	0.644	0.343	0.555	0.552	0.333
Qwen3.5-9B	`id`	0.975	0.972	0.573	0.572	0.333
Qwen3.5-9B	`depth`	0.799	0.821	0.480	0.478	0.333
Qwen3.5-9B	`lexical`	0.611	0.830	0.574	0.568	0.333

Table 2: **Position-matched post-trained *Qwen3.5* scoring comparison.** Answer-position probes, same-position unembedding, first-token vocabulary-head predictions, and complete sequence likelihood on the same 1,000-example controlled-logic splits. The large probe-vs.-sequence gap persists after instruction tuning: hidden states retain answer information that output likelihood fails to expose.

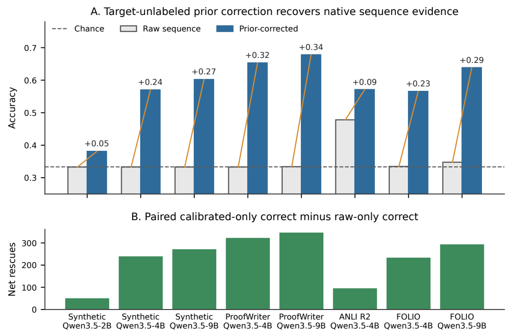

Figure 2: **Cross-task recovery from collapsed sequence scoring. (A)** Raw sequence accuracy (gray) vs. label-bias-corrected (blue); dashed line = chance (1/3). **(B)** Net per-example rescues (corrected minus broken). Offsets fitted on 25 unlabeled in-distribution scores.

with mean-normalized scores. FOLIO shows the same pattern on a separate 1,001-example held-out set, with per-example gains of +231 and +291 for *Qwen3.5-4B/9B*. Raw sequence likelihood, though collapsed, therefore preserves a useful relative ordering: corrected predictions fix more held-out examples than they break, and macro-F1 improves in every reported row.

Raw answer likelihood can therefore be a poor decision rule while preserving a useful ordering among candidate answers. The next question is whether the gains behave like a low-capacity bias correction or like a hidden task learner that exploits many in-domain examples.

4.4 Sample efficiency

Hypothesis. A true bias correction needs only enough examples to estimate two per-label offsets—25 to 100 should suffice for collapsed cases—while a model with already-high raw accuracy has little room to improve regardless of sample size. Table 4 and Figure 3 refit the same two-parameter correction on 25, 100, and 1,000 unlabeled in-domain examples. With only 25 examples, *Qwen3.5-2B/4B/9B* show positive accuracy gains across all 30 subsamples on the synthetic lexical split (mean accuracy 0.379/0.549/0.580); *Qwen3.5-4B/9B* match the pattern on ProofWriter (+0.216/+0.277). On ANLI R2, *Qwen3.5-2B/4B* both improve under 25-example offsets (+0.001/+0.054 mean gain), while *Qwen3.5-9B* ANLI serves as the expected control: raw sequence accuracy is already 0.638, leaving no room to improve. Sample efficiency therefore matches a simple bias correction: all 30 out of 30 subsamples yield positive gains at 25 examples on collapsed rows, and accuracy changes by less than 0.01 moving from 25 to 1,000 examples—consistent with estimating two parameters, not learning a task.

4.5 Shortcut and label-count controls

Hypothesis. Two alternative explanations could undermine the recovery results: the correction may exploit lexical shortcuts, or it may only improve the overall label distribution without fixing the right examples. A clean result should remain positive on TF-IDF-missed examples *and* exceed label-count permutation baselines on the same slice. On ProofWriter TF-IDF-missed examples, the 25-example correction gives *Qwen3.5-4B/9B* accuracies 0.622/0.643 with gaps above the permutation baseline of +0.287/+0.305; *Qwen3.5-4B* also remains positive on ANLI R2, reaching 0.557 with a +0.226 gap above baseline. Table 5 collects the main controls (full rows in Appendix Table 27); a result counts as clean only when it passes every

Task	Model	N	Raw	Cal.	Paired Gain	Macro-F1	p
synthetic lexical	*Qwen3.5-2B*	1000	0.333	0.381	272-224=+48	0.371	$p = 0.035$
	Qwen3.5-4B	1000	0.333	0.570	351-114=+237	0.572	$p < .001$
	Qwen3.5-9B	1000	0.333	0.602	371-102=+269	0.591	$p < .001$
ProofWriter	*Qwen3.5-4B*	1000	0.333	0.653	478-158=+320	0.650	$p < .001$
	Qwen3.5-9B	1000	0.334	0.678	474-130=+344	0.678	$p < .001$
ANLI R2	*Qwen3.5-4B*	1000	0.478	0.571	149-56=+93	0.573	$p < .001$
FOLIO diagnostic	*Qwen3.5-4B*	1001	0.335	0.565	407-176=+231	0.553	$p < .001$
	Qwen3.5-9B	1001	0.348	0.638	437-146=+291	0.632	$p < .001$

Table 3: **Sequence scores preserve answer evidence after label-bias correction.** Additive per-label offsets are fit on unlabeled in-domain score distributions and frozen before held-out evaluation. Per-example gain reports examples the corrected rule gets right that the raw rule gets wrong, minus the reverse.

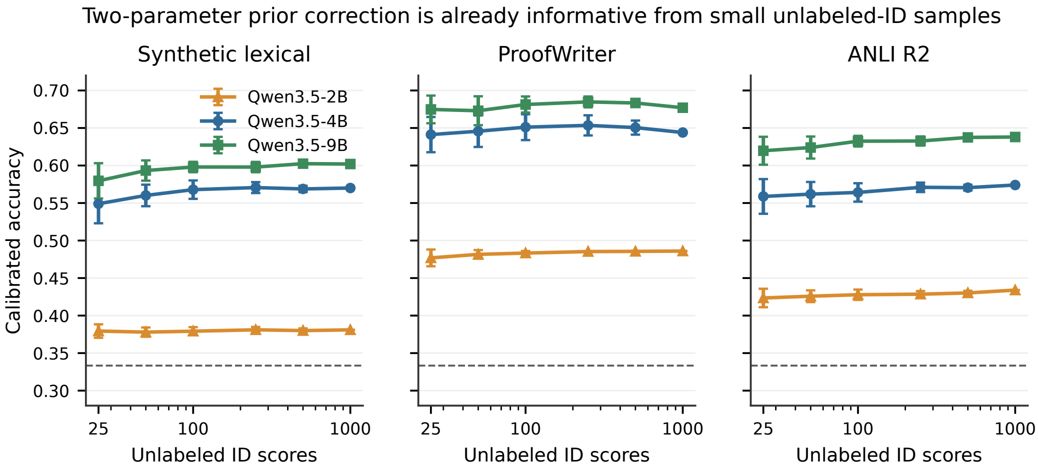

Figure 3: **25 unlabeled examples suffice for the main collapsed *Qwen3.5* rows.** Calibrated accuracy (y-axis) vs. number of unlabeled ID scores used to fit the correction (x-axis, log-scale); error bars show seed variance. *Qwen3.5-9B* ANLI, already near ceiling, serves as a scope-delimiting control.

paired check.

Control	Alternative explanation	Main outcome
TF-IDF-missed examples	Lexical shortcuts.	*Qwen3.5-4B/9B* stay positive on synthetic and ProofWriter hard examples; *Qwen3.5-4B* also on ANLI R2.
Label-count permutation baselines	Label-distribution matching only.	Main *Qwen3.5* rows exceed their permutation baselines; multiseed *Qwen3.5-9B* lexical-OOD gap is $+0.205$.
Prior perturbations	Sensitive to the assumed prior.	Worst-case gains remain positive on synthetic ($+0.193/+0.242$) and ProofWriter ($+0.281/+0.294$).
25-example fits	Requires large calibration set.	Collapsed rows recover from 25 unlabeled examples on synthetic, ProofWriter, and ANLI R2.
Surface/model scope	Answer-string artifact or universal effect.	20 prompt/verbalizer variants and equal-token ABC preserve collapse; *Pythia* and high-raw *Qwen3.5* rows delimit scope.

Table 5: **Control summary.** Corrected decisions are checked against lexical filters, label-count permutation baselines, prior perturbations, small-sample fits, surface changes, and model-scope controls.

The same table also covers prior and surface robustness: worst-case gains under prior perturbation remain positive, and 20 prompt/verbalizer variants plus equal-token ABC controls preserve score collapse rather than removing it. Across three independent 1,000-example lexical-OOD seeds, *Qwen3.5-9B* stays positive on TF-IDF-missed examples (0.574 ± 0.025, minimum gain $+0.086$, gap above permutation baseline $+0.205$). On FOLIO, *Qwen3.5-4B/9B* reach 0.550/0.624 on the 614 TF-IDF-missed validation examples. Full tables are in Appendix Tables 21, 30, 12, 11, 28, and 46.

4.6 Complementarity with source-task transfer

Hypothesis. Corrected sequence scores should add information beyond models trained on other tasks; otherwise their evidence is redundant with cross-task transfer. ProofWriter isolates this test because source-task adapters are selected without ProofWriter labels and frozen before any target sequence scores are used. The *Qwen4+9+Llama* source ensemble reaches 0.700 ProofWriter OOD accuracy and 0.682 on TF-IDF-missed examples;

Task	Model	Raw Acc.	25 ID	100 ID	1000 ID	Pos@25	Macro-F1@25
Synthetic lexical	*Qwen3.5-2B*	0.333	0.379±0.009 (+0.030)	0.379±0.005 (+0.036)	0.381±0.000 (+0.048)	30/30	0.363
Synthetic lexical	*Qwen3.5-4B*	0.333	0.549±0.026 (+0.142)	0.568±0.012 (+0.195)	0.570±0.000 (+0.237)	30/30	0.546
Synthetic lexical	*Qwen3.5-9B*	0.333	0.580±0.024 (+0.180)	0.598±0.007 (+0.248)	0.602±0.000 (+0.269)	30/30	0.571
ProofWriter	*Qwen3.5-4B*	0.333	0.641±0.023 (+0.216)	0.651±0.017 (+0.289)	0.644±0.000 (+0.311)	30/30	0.638
ProofWriter	*Qwen3.5-9B*	0.334	0.675±0.018 (+0.277)	0.681±0.011 (+0.314)	0.677±0.000 (+0.343)	30/30	0.674
ANLI R2	*Qwen3.5-2B*	0.389	0.423±0.012 (+0.001)	0.428±0.007 (+0.020)	0.434±0.000 (+0.045)	30/30	0.418
ANLI R2	*Qwen3.5-4B*	0.445	0.559±0.023 (+0.054)	0.564±0.012 (+0.083)	0.574±0.000 (+0.129)	30/30	0.554
ANLI R2	*Qwen3.5-9B*	0.638	0.620±0.019 (-0.068)	0.632±0.007 (-0.021)	0.638±0.000 (+0.000)	6/30	0.615

Table 4: **Sample efficiency of the unsupervised label-bias correction.** Per-label offsets are fit from small unlabeled samples and evaluated on the full held-out set; consistent gains at 25 examples rule out explanations requiring large in-domain calibration sets.

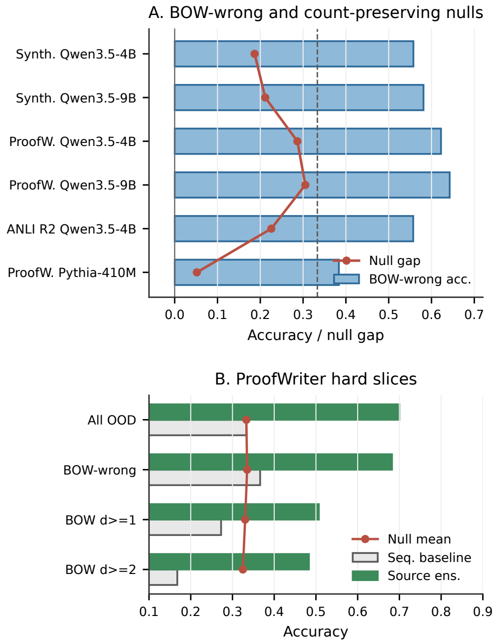

Figure 4: **Shortcut and label-count controls for TF-IDF-missed examples.** **(A)** Bars: calibrated accuracy on examples a TF-IDF classifier misses; red dots: gap above label-count permutation nulls. *Qwen3.5* rows exceed both chance and nulls; *Pythia* does not. **(B)** ProofWriter hard slices: source ensemble (green) exceeds the sequence baseline (gray) and null means (red) across depth strata.

unsupervised combination with the corrected sequence scores lifts the ensemble to 0.720/0.707, with 20 net per-example gains over the source-only baseline. Corrected sequence scores therefore add target-domain evidence that cross-task transfer does not already capture, and the member ablations in Appendix Table 32 show that the complementarity is selective rather than driven by a single source.

5 Discussion

Evidence is conjunctive. High probe accuracy shows only that the answer is decodable from hidden states; it does not show that the model's own scoring recovers it. A positive per-example gain that does not exceed the label-count permutation baseline reflects label-distribution improvement, not per-example recovery. The main *Qwen3.5* results pass every step; *Pythia* rows that improve raw accuracy but fail the permutation baseline, and high-raw *Qwen3.5-9B* on ANLI, delimit where this conclusion does *not* hold.

Mechanism. The scale pattern fits the additive decomposition $s_y(x) = \tilde{s}_y(x) + b_y$: in larger models the task signal $\tilde{s}_y(x)$ grows more discriminative while per-label offsets b_y absorbed from pretraining simultaneously grow, collapsing absolute scores while relative orderings survive. Two parameters restore the ordering without target labels; the boundary conditions (high-raw rows, *Pythia* failures) are exactly what this account predicts.

Practical implications. Model comparisons based solely on raw likelihood scoring risk underestimating representational capacity. Adding a label-bias check before reporting negative reasoning results is inexpensive: fitting two offsets from 25 unlabeled examples costs negligible compute. The probe–sequence–correction comparison can serve as a lightweight diagnostic layer on top of existing benchmark pipelines.

6 Conclusion

We have shown that collapsed candidate-answer likelihood can be a scoring artifact rather than evidence of absent reasoning. A two-parameter label-bias correction, fitted from 25 unlabeled examples, recovers significant per-example accuracy on controlled logic, ProofWriter, ANLI, and FOLIO—gains that survive lexical-shortcut filters and label-count permutation baselines. Scope controls identify where the recovery does *not* hold: *Pythia* models and already-high-accuracy *Qwen3.5* rows serve as boundary conditions rather than successes. For evaluation practice, future reasoning assessments should verify claimed deficits against label-frequency, shortcut, and permutation controls before attributing them to absent capacity. Readout is an underexplored lever complementary to scaling.

Limitations

The study is synthetic-first; external datasets serve as pressure tests for generality rather than proof of a universal mechanism. FOLIO is treated as a diagnostic case because the correction is fit on a 203-example validation set and evaluated on a separate 1,001-example held-out set. The correction depends on an assumed label distribution: prior perturbation experiments show local robustness for the main *Qwen3.5* rows but also identify scope cases, so the sample-efficient result is evidence about controlled score offsets rather than a deployment guarantee.

The method is not a causal circuit analysis. Probes, vocabulary-head predictions, and sequence scores identify *where* scoring and representation diverge, without identifying the circuit responsible. Component interventions and activation-patching experiments are excluded from the main claims because they do not provide equally clean full-split repair evidence. The broader artifact suite required roughly 400 H100 GPU-hours; this paper uses a selected subset of those results.

Finally, the comparison assumes a fixed, enumerable set of candidate labels. Extending it to open-ended generation would require defining candidate-answer strings, a target label distribution, and matched shortcut and label-count controls.

Ethics Statement

This is an interpretability and evaluation study using synthetic data, public benchmarks, and public model checkpoints. It does not collect personal data or introduce a deployed decision system. The chief ethical risk is overclaiming: corrected scores or probes could be misread as proof that a model reasons correctly internally. We therefore report scope controls, shortcut audits, and explicit limits, and we treat correction as diagnostic evidence rather than as a safety or reliability guarantee.

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A Appendix Overview

Appendix blocks serve two roles: *parallel results* (same method, additional domain or variant) and *setup details* (implementation knobs and reproducibility artifacts). Table 6 indexes each main-text claim to the appendix tables a reviewer should consult; Table 7 maps the full content by block.

B Data, Models, and Reproducibility

This section stores setup details that would interrupt the main narrative: split sizes, score types, offset-fitting rules, uncertainty procedures, and checkpoint revisions. All main held-out evaluations use at least 1,000 examples except the explicitly marked FOLIO diagnostic fit surface. The controlled-logic data are generated from fixed scripts and seeds; all retained result summaries and figure-generation scripts are included in the supplementary material. Table 8 makes the paper-facing rerun protocol visible in the PDF.

The main text describes the measurement logic rather than every implementation knob. In code, additive prior offsets are selected by deterministic two-dimensional search after fixing one label as the zero offset, ranking candidates by prior fit, lower collapse, and smaller norm. Bootstrap intervals, paired tests, sampled-ID repeats, TF-IDF baselines, and probe settings are documented in the supplementary material; they affect uncertainty reporting and robustness checks, not the high-level order of score, fit offsets without labels, freeze, and evaluate held out.

Controlled-logic dataset. The controlled-logic benchmark is a balanced three-way deduction task (`true`/`false`/`unknown`) generated from fixed scripts and seeds. Each example presents shuffled facts and rules over nonce terms (e.g. *dax*, *wug*, *mip*) and asks whether a named entity has a query property. Labels are perfectly balanced (1/3 each) and distractors are label-balanced: for every label, two positive-property and two negative-property rules mention the query property, so that a model must perform multi-hop deduction rather than exploit shallow lexical cues.

Example (depth-2, label `true`*):*

> **Facts and rules:** Ivo is a marn. Every marn is a pavo. Every joto is bright. No wug is bright. Every pavo is a joto. Every joto is fragile. No dax is bright. Every sarn is bright.
> **Question:** Is Ivo bright?
> **Answer:** true (chain: marn $\rightarrow$ pavo $\rightarrow$ joto $\rightarrow$ bright)

The splits test different generalization axes:

- **ID test** (1,000 examples): same depth distribution (1–2 hops) and template style as the fit set.
- **Depth OOD** (1,000): deeper chains (3–5 hops), same templates.
- **Lexical OOD** (1,000): same depth (1–2) but paraphrased templates (e.g. "*All {src} things are {dst} things*", "*{name} belongs to the {term} class*").
- **Stress/counterfactual**: additional generator seeds or counterfactual rule inversions for robustness checks.

All splits share the same nonce vocabulary and name set; only depth, template variation, and seed differ across conditions.

The model manifest records checkpoint status and revisions. It is a reproducibility artifact: base and post-trained checkpoints are different experimental conditions, so model-family comparisons should not be reduced to a single model-size story.

Main-text claim	Verify in
Probe accuracy exceeds sequence scoring (§4.2)	Tabs. 13, 18
Unsupervised correction recovers per-example gains (§4.3)	Tabs. 20, 29, 42
25 unlabeled examples suffice (§4.4)	Tabs. 36, 43
Gains survive TF-IDF-missed examples (§4.5)	Tabs. 27, 28, 33
Gains exceed permutation baseline (§4.5)	Tabs. 23, 31, 41
Sequence scores complement source transfer (§4.6)	Tabs. 32, 33

Table 6: **Claim verification index.** For each main-text claim, the right column lists the appendix tables a reviewer should consult to verify it.

Appendix block	Type	Content	Key tables
Data, models, reproducibility	*Setup details*	Splits, fit/eval sizes, score types, offset rules, uncertainty procedures, and checkpoint revisions.	Tabs. 8, 9
Surface and verbalizers	*Setup details*	Prerequisite surface-leakage, equal-token, and prompt/verbalizer controls that guard against metadata and length artifacts.	Tabs. 10–12
Expanded readout audit	*Parallel results* (§4.2)	Base-model position-matched results, vocabulary-head synthesis, scoring sensitivity, and probe scope controls.	Tabs. 14–18
Controlled-logic parallel results	*Parallel results* (§4.3–§4.5)	Same correction on additional seeds, normalization variants, TF-IDF-missed slices, and label-count nulls.	Tabs. 20–27
ProofWriter parallel results	*Parallel results* (§4.6)	Natural-language logic results and source/sequence-score complementarity on OOD and hard slices.	Tabs. 29–38
ANLI and FOLIO parallel results	*Parallel results* (§4.3)	External-domain diagnostics for adversarial NLI and natural-language first-order logic.	Tabs. 39–46
Auxiliary scope controls	*Setup details*	Component ablations, intervention audits, model-family controls, and large adapter grids that bound the main interpretation.	Tab. 47

Table 7: **Appendix organization.** Each block is either *parallel results* (same method, different domain or variant, cross-referenced to the main section) or *setup details* (implementation knobs, controls, and reproducibility artifacts that keep the main text uncluttered).

Dataset / split	Fit source	Fit n	Eval n	Score type	Offset fitting	Uncertainty / nulls	Source script and tables
Controlled logic: ID, depth, lexical, stress, counterfactual	Synthetic ID score rows	1,000	1,000 per split	Summed sequence log likelihood; mean-score robustness	Two additive label biases; balanced three-way prior; grid step 0.05, span 12	10,000 bootstraps; 10,000 label-count permutations for null rows	`synthetic_sequence_prior_calibration.py`; Tables 20–23
Controlled lexical sample efficiency and TF-IDF-missed slices	Synthetic ID subsamples	25–1,000	1,000 lexical OOD; TF-IDF-missed subset recomputed	Summed sequence log likelihood	Same two-bias fit on deterministic subsamples	30 repeats for sampled fits; 10,000 permutations for TF-IDF-missed/null rows	`native_prior_sample_efficiency.py`; `native_prior_bow_sample_efficiency.py`; Tables 4, 27
ProofWriter sequence-score OOD	ProofWriter ID score rows	1,000	1,000 OOD	Mean-normalized sequence scores for retained rows; summed-score robustness	Two additive label biases; balanced prior; grid step 0.05, span 12	10,000 bootstraps; 10,000 label-count permutations	`proofwriter_native_prior_calibration.py`; Tables 29–31
ProofWriter source/sequence-score hybrid	ProofWriter ID score rows; source adapters fixed before target evaluation	1,000, with 25–1,000 subsamples for efficiency rows	1,000 OOD; hard slices	Source-adapter scores plus sequence scores	Label-free source/sequence-score fusion; score weight and prior sensitivity reported separately	10,000 bootstraps and permutations; 30 repeats for sampled-ID rows	`proofwriter_source_native_hybrid_ensemble.py`; Tables 32–37
ANLI R2/OOD sequence-score and hybrid diagnostics	ANLI R1/ID score rows	1,000	1,000 R2/OOD	Summed sequence scores; mean-normalized sequence scores in hybrid diagnostics	Two additive label biases; balanced ANLI prior; grid step 0.05, span 5	10,000 bootstraps or permutations; 30 repeats for sampled-ID hybrid rows	`anli_sequence_prior_calibration.py`; `anli_native_adapter_hybrid_sample_efficiency.py`; Tables 39–41
FOLIO diagnostic	FOLIO `id_test` validation score rows	203	1,001 prepared train examples	Summed and mean-normalized sequence scores	Two additive label biases; balanced prior; grid step 0.05, span 12	10,000 bootstraps; 10,000 label-count permutations; 30 repeats for smaller fit budgets	`folio_native_prior_calibration.py`; Tables 42–46

Table 8: **Protocol-level reproducibility map.** The table lists the unlabeled fit surface, held-out evaluation surface, score type, offset-fitting rule, uncertainty or null procedure, and source script for the main appendix blocks. FOLIO is explicitly diagnostic: the offset fit uses 203 validation examples and evaluates on a separate 1,001-example prepared train surface.

Model	Status	Revision	Trust RC
Qwen3.5-0.8B	post-trained	2fc06364	True
Qwen3.5-2B	post-trained	15852e8c	True
Qwen3.5-4B	post-trained	851bf6e8	True
Qwen3.5-9B	post-trained	c2022362	True
Qwen3.5-0.8B-Base	base-pretrained	dc7cdfe2	True
Qwen3.5-2B-Base	base-pretrained	b1485b2f	True
Qwen3.5-4B-Base	base-pretrained	1001bb4d	True
Qwen3.5-9B-Base	base-pretrained	68c46c4b	True
Pythia-160M	base-pretrained	582159a2	False
Pythia-410M	base-pretrained	c4fc8d58	False
Pythia-1B	base-pretrained	7199d8fc	False
Pythia-1.4B	base-pretrained	554d9c1b	False
Pythia-2.8B	base-pretrained	7d977fed	False
Pythia-6.9B	base-pretrained	372b1c08	False
Pythia-12B	base-pretrained	39c1bd94	False

Table 9: **Model manifest.** The retained artifacts record checkpoint status and revisions for the principal *Qwen3.5* and *Pythia* models, making base and post-trained comparisons auditable rather than a model-size-only story.

C Surface and Verbalizer Controls

Before reading the sequence-score recovery tables, the surface controls check what shallow information is available in the experimental prompts. The surface-leakage table asks whether controlled-logic surfaces are solvable by metadata or template features. Some metadata baselines are strong on structured controlled surfaces, which is why the main evidence relies on TF-IDF-missed and permutation checks alongside raw accuracy.

Feature group	ID	Depth	Lexical	Stress
Full text	0.640	0.550	0.381	0.345
Query only	0.353	0.347	0.326	0.333
Facts only	0.631	0.564	0.389	0.326
Prop-redacted	0.674	0.682	0.422	0.336
Template only	0.673	0.604	0.368	0.330
Metadata no polarity	0.352	0.333	0.352	0.320
Metadata all	1.000	1.000	1.000	1.000

Table 10: **Surface leakage audit.** TF-IDF and metadata baselines test whether controlled-logic prompts leak labels through shallow features; strong metadata rows motivate the TF-IDF-missed and permutation controls used for the main claim.

The prompt and verbalizer controls address a different concern: LM sequence collapse might be an artifact of answer strings, prompt wording, or verbalizer-length effects. These tables establish that collapse persists after controlling for surface form, motivating the label-bias correction that follows.

D Expanded Readout Audit

This block stores expanded diagnostics for the readout gap summarized in the main text. Table 13 gives the full position-matched results for *Qwen3.5* Base, which show the same probe-vs.-sequence gap as the post-trained models in the main text. Table 14 compresses the same message across *Qwen3.5* Base sizes: answer-position probes remain substantially above the best vocabulary-head direction and complete sequence likelihood, even when the strongest vocabulary-head position is selected. Table 15 further tests whether sum vs. mean normalization explains the collapse; the gap persists under both.

Model	Plain Seq.	ABC Seq.	ABC Options Seq.
Pythia-410M	0.332 (+0.208)	0.333 (+0.207)	0.344 (+0.196)
Pythia-12B	0.331 (+0.307)	0.333 (+0.305)	0.329 (+0.309)
Qwen3.5-0.8B	0.322 (+0.224)	0.333 (+0.213)	0.334 (+0.212)
Qwen3.5-9B	0.355 (+0.482)	0.333 (+0.504)	0.496 (+0.341)

Table 11: **Equal-token verbalizer audit.** Equal-token answer strings hold verbalizer length fixed; remaining sequence-score collapse shows that raw sequence scores are not explained solely by answer-token length.

The following probe tables control scope: selectivity, direction stability, gap significance, and training-size sensitivity. Table 16 fits probes on random or auxiliary labels to bound what accuracy can mean; Table 17 transfers learned directions across splits to test distribution sensitivity.

Tables 18 and 19 close the probe audit: the first quantifies the probe/sequence gap with bootstrap intervals and paired tests; the second checks whether probe accuracy is sensitive to training-set size.

E Controlled-Logic Parallel Results and Label-Bias Details

Parallel controlled-logic results extending §4.3–§4.5: the same correction applied to additional seeds, score normalizations, TF-IDF-missed slices, and label-count null comparisons. *Pythia-2.8B/12B* rows appear throughout as model-family scope controls.

Normalization and permutation nulls. The next two tables check whether the controlled-split result can be explained away by score normalization or label-count effects. Table 22 uses mean-normalized candidate scores, while Table 23 compares the corrected predictions against label-count permutations and shift controls. These checks test whether corrected predictions are aligned with the right held-out examples, beyond reporting accuracy alone.

Table 24 repeats the correction on independently generated lexical-OOD seeds to confirm the effect is not tied to one data sample.

TF-IDF-missed slices. The TF-IDF-missed tables are the strictest surface-artifact checks in the

Model	Variants	Slot Mean	Slot Max	Seq. Mean	Seq. Max	Seq. Collapse
Pythia-410M	20	0.336	0.366	0.339	0.354	0.764
Pythia-12B	20	0.333	0.342	0.335	0.380	0.895
Qwen3.5-0.8B	20	0.335	0.351	0.332	0.336	0.962
Qwen3.5-9B	20	0.402	0.567	0.383	0.563	0.886
Qwen3.5-0.8B-Base	20	0.341	0.380	0.339	0.373	0.964
Qwen3.5-9B-Base	20	0.448	0.594	0.392	0.551	0.873

Table 12: **Prompt/verbalizer robustness.** Prompt and verbalizer variants stress-test whether sequence-score collapse is a surface artifact; recovery is evaluated with the label-bias correction rather than a single wording.

Model	Split	Prompt Probe	Answer Probe	Slot W_U	Vocab Head	Seq.
Qwen3.5-0.8B-Base	`id`	0.855	0.866	0.333	0.333	0.333
Qwen3.5-0.8B-Base	`depth`	0.603	0.655	0.333	0.333	0.333
Qwen3.5-0.8B-Base	`lexical`	0.607	0.655	0.333	0.333	0.333
Qwen3.5-2B-Base	`id`	0.915	0.933	0.333	0.333	0.333
Qwen3.5-2B-Base	`depth`	0.679	0.640	0.333	0.333	0.333
Qwen3.5-2B-Base	`lexical`	0.559	0.653	0.333	0.333	0.333
Qwen3.5-4B-Base	`id`	0.958	0.968	0.366	0.372	0.336
Qwen3.5-4B-Base	`depth`	0.741	0.860	0.339	0.340	0.333
Qwen3.5-4B-Base	`lexical`	0.626	0.751	0.347	0.349	0.333
Qwen3.5-9B-Base	`id`	0.968	0.975	0.627	0.617	0.374
Qwen3.5-9B-Base	`depth`	0.791	0.853	0.548	0.538	0.336
Qwen3.5-9B-Base	`lexical`	0.610	0.873	0.607	0.604	0.361

Table 13: **Position-matched *Qwen3.5* Base scoring comparison.** Answer-position probes, same-position unembedding, first-token vocabulary-head predictions, and complete sequence likelihood on the same 1,000-example controlled-logic splits. The probe-vs.-sequence gap is consistent with the post-trained results reported in the main text.

Model	Slot Probe	Best W_U	Seq.	ΔPeak	ΔFinal
Qwen3.5-0.8B-Base	0.639	0.334	0.333	+0.305	+0.321
Qwen3.5-2B-Base	0.691	0.357	0.333	+0.334	+0.296
Qwen3.5-4B-Base	0.846	0.515	0.333	+0.331	+0.236
Qwen3.5-9B-Base	0.910	0.607	0.361	+0.303	+0.266

Table 14: ***Qwen3.5* Base vocabulary-head synthesis.** Selecting the best vocabulary-head position narrows the probe/vocabulary-head gap, but answer-position probes remain higher and full-answer sequence likelihood remains the weakest scoring method.

Model	Answer	Random	Depth	Query	TF-IDF-easy
Pythia-410M	0.564 (+0.230)	0.341 (+0.007)	0.739 (+0.229)	0.726 (+0.592)	0.380 (-0.239)
Pythia-12B	0.618 (+0.284)	0.338 (+0.004)	0.687 (+0.177)	0.723 (+0.589)	0.497 (-0.122)
Qwen3.5-0.8B	0.600 (+0.266)	0.304 (-0.030)	0.752 (+0.242)	0.626 (+0.492)	0.538 (-0.081)
Qwen3.5-9B	0.651 (+0.317)	0.315 (-0.019)	0.639 (+0.129)	0.882 (+0.748)	0.548 (-0.071)

Table 16: **Probe selectivity controls.** Random-label and auxiliary-target fits separate answer evidence from arbitrary-label or side-feature decodability, bounding what probe accuracy can support before sequence-score tests.

Model	Layer	Cosine	Canon.→CF	CF→Lex.	Agree Lex.
Pythia-410M	peak	0.177	0.458	0.386	0.178
	final	0.065	0.348	0.343	0.343
Qwen3.5-0.8B	peak	0.040	0.465	0.508	0.694
	final	0.316	0.358	0.388	0.456

Table 17: **Probe direction stability.** Direction-transfer diagnostics train and evaluate answer directions across splits; uneven transfer shows that probe directions are distribution-sensitive diagnostic upper bounds.

Model	Sum Acc	Sum majority	Mean Acc	Mean majority
Pythia-2.8B	0.329	`true` (0.947)	0.329	`true` (0.947)
Pythia-12B	0.334	`true` (0.852)	0.334	`true` (0.852)
Qwen3.5-0.8B	0.333	`unknown` (1.000)	0.333	`unknown` (1.000)
Qwen3.5-0.8B-Base	0.333	`unknown` (1.000)	0.333	`unknown` (1.000)
Qwen3.5-9B	0.333	`unknown` (0.999)	0.333	`unknown` (0.999)
Qwen3.5-9B-Base	0.361	`unknown` (0.972)	0.361	`unknown` (0.972)

Table 15: **Sequence scoring sensitivity.** Sum/mean sequence scoring tests whether candidate length or normalization explains complete-sequence collapse; the probe/sequence gap persists.

sequence-score section. They first remove examples solved by a TF-IDF baseline and then ask whether the corrected predictions still beat a label-count permutation baseline. This is the closest appendix analogue of the main text's shortcut control: it separates recovered sequence-score evidence from lexical shortcuts that a shallow model already captured.

F ProofWriter Sequence Scores and Source-Task Evidence

Parallel results extending §4.3 and §4.6 to natural-language proofs. The first block (Tables 29–31) repeats the sequence-score audit on ProofWriter: mean-normalized scores test whether recovery is an artifact of candidate-string length; prior sensitiv-

Model	Probe–W_U	Probe–Seq.	Vocab–Seq.
Pythia-410M	+0.121 [+0.083, +0.160]; $p < .001$	+0.121 [+0.083, +0.160]; $p < .001$	+0.000 [+0.000, +0.000]; $p = 1.000$
Pythia-12B	+0.137 [+0.091, +0.183]; $p < .001$	+0.132 [+0.086, +0.177]; $p < .001$	-0.004 [-0.025, +0.016]; $p = 0.769$
Qwen3.5-0.8B	+0.325 [+0.291, +0.358]; $p < .001$	+0.325 [+0.291, +0.358]; $p < .001$	+0.000 [+0.000, +0.000]; $p = 1.000$
Qwen3.5-9B	+0.247 [+0.207, +0.288]; $p < .001$	+0.488 [+0.452, +0.523]; $p < .001$	+0.237 [+0.180, +0.293]; $p < .001$
Qwen3.5-0.8B-Base	+0.322 [+0.281, +0.363]; $p < .001$	+0.322 [+0.281, +0.363]; $p < .001$	+0.000 [+0.000, +0.000]; $p = 1.000$
Qwen3.5-9B-Base	+0.266 [+0.234, +0.299]; $p < .001$	+0.512 [+0.475, +0.550]; $p < .001$	+0.243 [+0.194, +0.292]; $p < .001$

Table 18: **Probe/sequence gap significance.** Bootstrap intervals and paired tests quantify probe/vocabulary-head/sequence gaps on matched examples; significant positive gaps show that decodable answer evidence exceeds the model's complete sequence likelihood scores.

Dataset	Model	100	250	500	1000	2000
Synth.	*Pythia-410M*	0.434 (+0.063)	0.489 (+0.065)	0.474 (+0.036)	0.545 (+0.112)	0.552 (+0.102)
	Pythia-12B	0.486 (+0.079)	0.532 (+0.119)	0.594 (+0.177)	0.636 (+0.207)	0.644 (+0.193)
	Qwen3.5-0.8B	0.480 (+0.039)	0.500 (-0.028)	0.499 (-0.044)	0.523 (-0.103)	0.530 (-0.109)
	Qwen3.5-9B	0.732 (+0.030)	0.836 (+0.059)	0.852 (+0.058)	0.910 (+0.060)	0.929 (+0.058)
ProofW.	*Pythia-410M*	0.465 (+0.114)	0.525 (+0.158)	0.558 (+0.174)	0.556 (+0.151)	0.587 (+0.164)
	Pythia-12B	0.502 (+0.098)	0.575 (+0.152)	0.611 (+0.154)	0.638 (+0.134)	0.652 (+0.126)
	Qwen3.5-0.8B	0.610 (+0.065)	0.633 (+0.023)	0.675 (+0.052)	0.680 (+0.030)	0.698 (+0.047)
	Qwen3.5-9B	0.748 (-0.003)	0.774 (+0.015)	0.791 (+0.014)	0.787 (+0.007)	0.799 (+0.015)

Table 19: **Probe training-size sensitivity.** Probe fits with increasing labeled examples check sample dependence; the pattern supports using probes as diagnostic upper bounds, with behavior evaluated by separate sequence-score tests.

Model	Split	Raw Acc.	Cal. Acc. [95% CI]	Δ [95% CI]	Cal. Macro-F1	Cal. Collapse
Pythia-2.8B	`lexical`	0.329	0.317 [0.288, 0.346]	-0.012 [-0.050, +0.026]	0.312	0.391
Pythia-12B	`lexical`	0.334	0.338 [0.309, 0.368]	+0.004 [-0.036, +0.044]	0.325	0.539
Qwen3.5-0.8B	`lexical`	0.333	0.369 [0.338, 0.399]	+0.036 [+0.000, +0.071]	0.357	0.489
Qwen3.5-2B	`lexical`	0.333	0.381 [0.351, 0.412]	+0.048 [+0.004, +0.092]	0.371	0.548
Qwen3.5-4B	`lexical`	0.333	0.570 [0.539, 0.601]	+0.237 [+0.197, +0.276]	0.572	0.453
Qwen3.5-9B	`lexical`	0.333	0.602 [0.571, 0.633]	+0.269 [+0.229, +0.310]	0.591	0.462
Qwen3.5-9B	`stress`	0.333	0.467 [0.436, 0.498]	+0.134 [+0.098, +0.170]	0.422	0.557
Qwen3.5-9B	`counterfactual`	0.333	0.515 [0.484, 0.546]	+0.182 [+0.142, +0.222]	0.479	0.485
Qwen3.5-0.8B-Base	`lexical`	0.333	0.417 [0.386, 0.447]	+0.084 [+0.047, +0.120]	0.407	0.483
Qwen3.5-9B-Base	`lexical`	0.361	0.586 [0.556, 0.617]	+0.225 [+0.189, +0.261]	0.566	0.558

Table 20: **Controlled-logic label-bias correction.** Additive label offsets are fit from unlabeled in-domain scores and evaluated on held-out controlled-logic surfaces; larger *Qwen3.5* rows recover accuracy and macro-F1, while the *Pythia-2.8B/12B* rows provide model-family controls.

Model	Subset	N	Raw Acc.	Worst Acc.	Median Acc.	Worst Δ	Worst Macro-F1
Pythia-410M	`all`	1000	0.334	0.316	0.325	-0.018	0.312
Pythia-12B	`all`	1000	0.334	0.316	0.333	-0.018	0.288
Qwen3.5-0.8B	`all`	1000	0.333	0.369	0.373	+0.036	0.346
Qwen3.5-0.8B	`TF-IDF-missed`	645	0.468	0.397	0.408	-0.071	0.365
Qwen3.5-4B	`all`	1000	0.333	0.526	0.570	+0.193	0.529
Qwen3.5-4B	`TF-IDF-missed`	645	0.468	0.544	0.564	+0.076	0.535
Qwen3.5-9B	`all`	1000	0.333	0.575	0.594	+0.242	0.547
Qwen3.5-9B	`TF-IDF-missed`	645	0.468	0.538	0.589	+0.070	0.516
Qwen3.5-9B-Base	`all`	1000	0.361	0.541	0.572	+0.180	0.494
Qwen3.5-9B-Base	`TF-IDF-missed`	645	0.482	0.516	0.592	+0.034	0.481

Table 21: **Controlled-logic prior sensitivity.** Local target-prior perturbations stress-test the unlabeled offset fit; *Qwen3.5* rows keep positive worst-case deltas, supporting recovery beyond a finely chosen prior.

Model	Full Cal. / Δ	TF-IDF Cal. / Δ	TF-IDF Null Gap	Worst Full Δ	Worst TF-IDF Δ
Pythia-410M	0.329 / -0.005	0.319 / +0.098	-0.019 ($p = 0.8580$)	-0.012	+0.070
Pythia-12B	0.340 / +0.006	0.287 / +0.073	-0.029 ($p = 0.9653$)	-0.019	+0.056
Qwen3.5-0.8B	0.375 / +0.042	0.431 / -0.037	+0.038 ($p = 0.0155$)	+0.039	-0.078
Qwen3.5-4B	0.577 / +0.244	0.580 / +0.112	+0.204 ($p = 0.0001$)	+0.173	+0.047
Qwen3.5-9B	0.608 / +0.275	0.600 / +0.132	+0.235 ($p = 0.0001$)	+0.225	+0.048
Qwen3.5-9B-Base	0.591 / +0.230	0.605 / +0.122	+0.224 ($p = 0.0001$)	+0.174	+0.026

Table 22: **Mean-normalized controlled-logic robustness.** Mean-normalized candidate scores test length normalization directly; the main *Qwen3.5* pattern persists, so the summed-score recovery is not a candidate-length artifact.

Model	Split	Cal. Acc.	Null Acc. [95%]	Sum Obs.-Null	Mean Obs.-Null	Cal. Macro-F1	p_{perm}
Qwen3.5-4B	`stress`	0.452	0.333 [0.308, 0.358]	+0.119	+0.103	0.401	0.0001
Qwen3.5-4B	`counterfactual`	0.504	0.333 [0.306, 0.361]	+0.171	+0.156	0.484	0.0001
Qwen3.5-9B	`stress`	0.467	0.333 [0.307, 0.359]	+0.134	+0.136	0.422	0.0001
Qwen3.5-9B	`counterfactual`	0.515	0.333 [0.306, 0.361]	+0.182	+0.179	0.479	0.0001

Table 23: **Controlled-logic shift/permutation null.** Label-count permutation baselines and shift controls keep label counts fixed; positive permutation gaps indicate example-level recovery beyond a better label histogram.

Model	Score	Seeds	Raw Acc.	Cal. Acc.	Min Δ	Obs.-Null	Macro-F1
Pythia-410M	Sum	3	0.334±0.000	0.336±0.006	-0.004	+0.003	0.333
Pythia-12B	Sum	3	0.332±0.005	0.313±0.023	-0.045	-0.021	0.303
Qwen3.5-0.8B	Sum	3	0.333±0.000	0.377±0.020	+0.016	+0.044	0.370
Qwen3.5-9B	Sum	3	0.333±0.000	0.585±0.016	+0.230	+0.252	0.580
Qwen3.5-9B	Mean	3	0.333±0.000	0.586±0.019	+0.228	+0.253	0.582

Table 24: **Independent synthetic-seed label-bias correction.** New lexical-OOD seeds regenerate the evaluation and fit surfaces; *Qwen3.5-9B* remains positive across seeds, showing that the effect is not tied to one generator sample.

Model	Score	N	Raw Acc.	Worst Acc.	Median Acc.	Worst Δ	Worst Macro-F1
Pythia-410M	Sum	637±16	0.200	0.317	0.348	+0.106	0.306
Pythia-12B	Sum	637±16	0.206	0.281	0.300	+0.067	0.262
Qwen3.5-0.8B	Sum	637±16	0.468	0.362	0.384	-0.116	0.334
Qwen3.5-9B	Sum	637±16	0.468	0.529	0.571	+0.047	0.501
Qwen3.5-9B	Mean	637±16	0.468	0.524	0.568	+0.048	0.503

Table 25: **Independent synthetic-seed TF-IDF-missed sensitivity.** For each seed, the TF-IDF-missed subset and unlabeled offset fit are recomputed; positive *Qwen3.5* rows test robustness after lexical shortcut removal.

Model	Score	N	Worst Acc.	Null Acc.	Obs.-Null	Macro-F1	p_{perm}
Pythia-410M	Sum	637±16	0.317	0.327	-0.010	0.307	0.8848
Pythia-12B	Sum	637±16	0.281	0.317	-0.036	0.268	0.9999
Qwen3.5-0.8B	Sum	637±16	0.362	0.349	+0.013	0.334	0.9272
Qwen3.5-9B	Sum	637±16	0.529	0.343	+0.186	0.506	0.0001
Qwen3.5-9B	Mean	637±16	0.524	0.330	+0.194	0.512	0.0001

Table 26: **Independent synthetic-seed TF-IDF-missed nulls.** Null comparisons show whether seed-specific corrected predictions exceed label-count permutation baselines on the same TF-IDF-missed slices, separating example recovery from label counts.

ity and permutation tables bound the result against prior mismatch and label-count effects. The second block (Tables 32–38) tests whether label-free sequence scores add information beyond source-task transfer. Source adapters are fixed before any ProofWriter labels are seen, so any gain from adding sequence scores is genuinely target-domain evidence; member ablations confirm the complementarity is selective rather than driven by a single source model.

Like the controlled-logic and ANLI results, recovery on ProofWriter must clear a label-count permutation null before it can be credited as per-example separation rather than a beneficial shift in the recovered label histogram. The permutation null below shuffles the model's corrected predictions across OOD examples and asks whether the original corrected configuration outperforms the average shuffle: a positive permutation gap rules out the label-histogram explanation.

The remaining tables test complementarity: hard-slice audits, member ablations, weight sensitivity, sample efficiency, and prior sensitivity for the hybrid.

Table 35 varies the sequence-score weight after fixing the source ensemble, confirming a broad accuracy plateau rather than a single fragile fusion point. Tables 36 and 37 check sample efficiency and prior robustness; Table 38 ablates individual source members.

G ANLI and FOLIO Diagnostics

Parallel results extending §4.3 to two external domains. ANLI is an adversarial NLI stress test with less template-like examples; FOLIO is a natural-language first-order-logic diagnostic with a 203-

Task	Model	$N_{\text{TFIDF-miss}}$	Raw	25 ID	100 ID	1000 ID	NullGap@25	Pos@25
Synthetic lexical	*Pythia-410M*	645	0.222	0.321±0.028 (+0.036)	0.327±0.012 (+0.068)	0.332±0.000 (+0.110)	-0.018 (-0.034)	30/30
Synthetic lexical	*Qwen3.5-4B*	645	0.468	0.558±0.018 (+0.020)	0.563±0.010 (+0.062)	0.569±0.000 (+0.101)	+0.187 (+0.161)	30/30
Synthetic lexical	*Qwen3.5-9B*	645	0.468	0.581±0.030 (+0.031)	0.587±0.018 (+0.067)	0.594±0.000 (+0.126)	+0.212 (+0.167)	30/30
ProofWriter	*Pythia-410M*	576	0.321	0.384±0.009 (+0.042)	0.382±0.006 (+0.052)	0.384±0.000 (+0.062)	+0.052 (+0.030)	30/30
ProofWriter	*Qwen3.5-4B*	576	0.366	0.622±0.024 (+0.205)	0.619±0.018 (+0.212)	0.622±0.000 (+0.255)	+0.287 (+0.238)	30/30
ProofWriter	*Qwen3.5-9B*	576	0.366	0.643±0.024 (+0.217)	0.647±0.017 (+0.227)	0.651±0.000 (+0.285)	+0.305 (+0.253)	30/30
ANLI R2	*Pythia-410M*	660	0.283	0.307±0.022 (-0.005)	0.309±0.013 (-0.002)	0.309±0.000 (+0.026)	-0.014 (-0.026)	23/30
ANLI R2	*Qwen3.5-4B*	660	0.377	0.557±0.038 (+0.095)	0.580±0.018 (+0.162)	0.583±0.000 (+0.206)	+0.226 (+0.169)	30/30

Table 27: **TF-IDF-missed and label-count controls.** Offsets are fit without target labels and evaluated only on TF-IDF-missed examples; the observed-minus-null gap reports whether gains exceed a label-count permutation baseline on the same slice.

Model	Score	N	Raw Acc.	Cal. Acc.	Min Δ	Obs.-Null	Macro-F1
Pythia-410M	Sum	637±16	0.200	0.349±0.014	+0.133	-0.000	0.327
Pythia-12B	Sum	637±16	0.206	0.302±0.022	+0.086	-0.034	0.283
Qwen3.5-0.8B	Sum	637±16	0.468	0.386±0.033	-0.104	+0.033	0.361
Qwen3.5-9B	Sum	637±16	0.468	0.574±0.025	+0.086	+0.205	0.570
Qwen3.5-9B	Mean	637±16	0.468	0.579±0.023	+0.093	+0.208	0.575

Table 28: **Independent synthetic-seed TF-IDF-missed check.** The TF-IDF-missed subset and label-bias correction are rebuilt separately for seeds 52, 62, and 72; stable *Qwen3.5* rows show that shortcut-controlled recovery generalizes across generator seeds.

Audit	N	Raw Acc.	Cal./Min Acc.	Δ / Obs.-Null	Macro-F1
Mean sequence, OOD	1000	0.333	0.653 [0.623, 0.682]	+0.320 [+0.274, +0.366]	0.650
Prediction-count null	1000	0.333	0.653	+0.320 ($p = 0.0001$)	0.650
TF-IDF-missed stratum	576	0.366	0.627	+0.294 ($p = 0.0001$)	0.627
±0.10 prior sensitivity	1000	0.333	0.611 / 0.651	+0.278 worst	0.594 min

Table 29: **ProofWriter sequence-score mean robustness.** Unlabeled offsets fit on ProofWriter ID scores are evaluated on OOD mean-normalized sequence scores; *Qwen3.5* rows reproduce recovery on natural-language proofs.

Model	Raw Acc.	Min Acc.	Median Acc.	Min Macro-F1	Worst Δ Raw
Qwen3.5-0.8B	0.333	0.406	0.415	0.395	+0.073
Qwen3.5-2B	0.333	0.508	0.544	0.488	+0.175
Qwen3.5-4B	0.333	0.614	0.644	0.602	+0.281
Qwen3.5-9B	0.334	0.628	0.678	0.628	+0.294
OLMo-2-1B	0.362	0.533	0.561	0.528	+0.171
Llama-3.1-8B	0.333	0.458	0.475	0.454	+0.125
Pythia-410M	0.328	0.337	0.351	0.332	+0.009
Pythia-12B	0.343	0.322	0.335	0.322	-0.021

Table 30: **ProofWriter label-bias sensitivity.** Prior perturbations around the unlabeled fit report worst retained accuracy; positive *Qwen3.5* margins define the natural-language operating region.

Model	Cal. Acc.	Null Acc. [95%]	Obs.-Null	Cal. Macro-F1	p_{perm}
Qwen3.5-0.8B	0.415	0.333 [0.305, 0.362]	+0.082	0.415	0.0001
Qwen3.5-2B	0.559	0.334 [0.305, 0.363]	+0.225	0.556	0.0001
Qwen3.5-4B	0.644	0.333 [0.304, 0.363]	+0.311	0.645	0.0001
Qwen3.5-9B	0.677	0.333 [0.304, 0.363]	+0.344	0.677	0.0001
OLMo-2-1B	0.566	0.333 [0.305, 0.363]	+0.233	0.563	0.0001
Llama-3.1-8B	0.477	0.333 [0.304, 0.363]	+0.144	0.481	0.0001
Pythia-410M	0.353	0.334 [0.305, 0.363]	+0.019	0.353	0.1032
Pythia-12B	0.335	0.333 [0.305, 0.363]	+0.002	0.335	0.4625

Table 31: **ProofWriter label-count permutation.** Label-count permutations over the same OOD examples test whether recovered accuracy exceeds a favorable prediction histogram.

Fusion	Full Acc.	ΔSrc	TF-IDF Acc.	D$\geq$2 Acc.	MinR	Paired Gain	Null Gap
Source only	0.700	+0.000	0.682	0.483	0.613	0	+0.367
Source + all LM scores	0.720	+0.020	0.707	0.557	0.625	20	+0.387
Source + *OLMo-2-1B*	0.725	+0.025	0.720	0.577	0.619	25	+0.392
Source + *Qwen3.5-9B*	0.705	+0.005	0.674	0.450	0.628	5	+0.372
Source + *Qwen3.5-4B*	0.687	-0.013	0.670	0.463	0.607	-13	+0.354
Source + *Llama-3.1-8B*	0.682	-0.018	0.661	0.490	0.553	-18	+0.349
Source + *Qwen3.5-2B*	0.686	-0.014	0.670	0.497	0.598	-14	+0.353
Source + *Qwen3.5-0.8B*	0.686	-0.014	0.658	0.450	0.601	-14	+0.353
Source + *Pythia-12B*	0.694	-0.006	0.686	0.523	0.616	-6	+0.361
Source + *Pythia-410M*	0.691	-0.009	0.696	0.550	0.586	-9	+0.358

Table 32: **ProofWriter source-transfer plus sequence-score hybrid.** Source adapters are fixed without ProofWriter labels; label-free sequence-score correction tests whether sequence scores add evidence beyond source transfer.

Slice	N	Depth	Seq. Acc.	Cal. Acc.	ΔSeq.	MinR	Null Gap
All OOD	1000	0–5	0.333	0.700	+0.367	0.613	+0.367
TF-IDF-missed	576	0–5	0.366	0.682	+0.316	0.616	+0.347
TF-IDF depth 0	290	0	0.459	0.855	+0.397	0.737	+0.517
TF-IDF depth >=1	286	1–5	0.273	0.507	+0.234	0.410	+0.177
TF-IDF depth >=2	149	2–5	0.168	0.483	+0.315	0.400	+0.158

Table 33: **ProofWriter hard-slice audit for the source-task ensemble.** The strongest label-free row remains above the frozen sequence baseline and the label-count permutation baseline on TF-IDF-missed and higher-depth slices.

LM-score set	n_m	Full Acc.	ΔSrc	TF-IDF Acc.	D≥2 Acc.	D≥2 Δ	Paired Gain	D≥2 Null Gap
Source only	0	0.700	+0.000	0.682	0.483	+0.000	0	+0.158
All LM scores	8	0.720	+0.020	0.707	0.557	+0.074	20	+0.226
Drop *Pythia* family	6	0.702	+0.002	0.675	0.503	+0.020	2	+0.182
Drop *Qwen* family	4	0.694	-0.006	0.700	0.591	+0.107	-6	+0.259
Drop *OLMo* family	7	0.700	+0.000	0.689	0.510	+0.027	0	+0.187
Drop *Llama* family	7	0.711	+0.011	0.691	0.537	+0.054	11	+0.206
Drop *Pythia-410M*	7	0.717	+0.017	0.689	0.510	+0.027	17	+0.195
Drop *Pythia-12B*	7	0.712	+0.012	0.688	0.537	+0.054	12	+0.206
Drop *Qwen3.5-0.8B*	7	0.720	+0.020	0.710	0.577	+0.094	20	+0.248
Drop *Qwen3.5-2B*	7	0.708	+0.008	0.693	0.544	+0.060	8	+0.209
Drop *Qwen3.5-4B*	7	0.714	+0.014	0.701	0.557	+0.074	14	+0.222
Drop *Qwen3.5-9B*	7	0.715	+0.015	0.698	0.557	+0.074	15	+0.227
Drop *OLMo-2-1B*	7	0.700	+0.000	0.689	0.510	+0.027	0	+0.187
Drop *Llama-3.1-8B*	7	0.711	+0.011	0.691	0.537	+0.054	11	+0.206

Table 34: **ProofWriter source/sequence-score hybrid ablation.** With source selection fixed, dropping sequence-score families or members tests whether one component carries the gain; mixed effects show complementary but uneven sequence-score evidence.

w_{LM}	Full Acc.	ΔSrc	TF-IDF Acc.	D≥2 Acc.	MinR	Paired Gain	Null Gap
0	0.700	+0.000	0.682	0.483	0.613	0	+0.367
0.25	0.713	+0.013	0.698	0.530	0.622	13	+0.380
0.5	0.711	+0.011	0.698	0.544	0.619	11	+0.378
0.75	0.718	+0.018	0.705	0.564	0.625	18	+0.385
1	0.720	+0.020	0.707	0.557	0.625	20	+0.387
1.5	0.716	+0.016	0.700	0.557	0.610	16	+0.383
2	0.719	+0.019	0.701	0.557	0.613	19	+0.386
3	0.720	+0.020	0.703	0.570	0.604	20	+0.387

Table 35: **ProofWriter source/sequence-score weight sensitivity.** Sequence-score weights are varied after the source ensemble is fixed; a broad plateau rules out a single finely tuned fusion coefficient.

ID rows	Repeats	Full Acc.	Worst Full	TF-IDF Acc.	Worst TF-IDF	D≥2 Acc.	Max Coll.
25	30	0.698±0.018	0.659	0.679±0.020	0.639	0.535±0.048	0.489
50	30	0.701±0.016	0.663	0.681±0.017	0.637	0.548±0.050	0.556
100	30	0.705±0.013	0.679	0.687±0.014	0.660	0.542±0.037	0.421
250	30	0.714±0.007	0.697	0.698±0.009	0.672	0.553±0.026	0.409
500	30	0.717±0.004	0.706	0.702±0.004	0.693	0.557±0.012	0.374
1000	1	0.720±0.000	0.720	0.707±0.000	0.707	0.557±0.000	0.344

Table 36: **ProofWriter source/sequence-score sample efficiency.** Small unlabeled ProofWriter fit subsets are evaluated on OOD, TF-IDF-missed, and depth slices; stable rows show the hybrid remains effective with small unlabeled budgets.

Radius	Priors	Full Worst	TF-IDF Worst	D≥2 Worst	Full ΔSrc	D≥2 ΔSrc	D≥2 Null Gap
0.025	7	0.705	0.689	0.537	+0.005	+0.054	+0.208
0.050	19	0.690	0.672	0.483	-0.010	+0.000	+0.157
0.100	61	0.666	0.648	0.430	-0.034	-0.054	+0.142

Table 37: **ProofWriter source/sequence-score prior sensitivity.** Prior perturbations define how far the label-free hybrid travels under plausible target-prior shifts.

example fit surface. Both ask whether the same correction improves held-out accuracy, survives prior shifts, and beats a label-count permutation baseline.

ANLI. ANLI is included as an adversarial NLI pressure test rather than as a formal logic benchmark. The hybrid rows ask whether the same unlabeled score-offset idea remains stable when

Members	OOD Acc./F1	OOD MinR	TF-IDF Acc./F1	TF-IDF MinR	TF-IDF Coll.	TF-IDF Null Gap
Qwen4	0.665/0.665	0.595	0.635/0.637	0.578	0.352	+0.301
Qwen9	0.695/0.695	0.625	0.661/0.664	0.578	0.345	+0.327
Llama	0.472/0.473	0.393	0.476/0.476	0.403	0.358	+0.142
Qwen4+Qwen9	0.688/0.687	0.595	0.653/0.655	0.569	0.340	+0.319
Qwen4+Llama	0.658/0.657	0.556	0.656/0.659	0.583	0.354	+0.321
Qwen9+Llama	0.676/0.675	0.568	0.674/0.675	0.578	0.335	+0.340
Qwen4+Qwen9+Llama	0.700/0.700	0.613	0.682/0.685	0.616	0.361	+0.347

Table 38: **ProofWriter source-ensemble member ablation.** Multiple source-member subsets retain gains, so the source side is not a one-adapter artifact.

the surface distribution is less template-like. The sample-efficiency and prior-sensitivity tables therefore matter more than a single best accuracy: they show whether the diagnostic survives small unlabeled budgets and local prior changes.

FOLIO. FOLIO is closer to the target phenomenon because it is natural-language first-order logic, but the available fit surface is small. We therefore label the FOLIO results as diagnostic. The fit uses the 203-example validation surface and evaluates on a separate prepared 1,001-example train surface; this split is a stress test for the audit rather than a standard leaderboard estimate.

Table 43 checks whether the diagnostic result degrades under smaller unlabeled budgets. Tables 44 and 45 then test whether the corrected accuracy exceeds its label-count permutation baseline and remains stable under prior perturbations. Finally, Table 46 restricts evaluation to TF-IDF-missed examples.

H Auxiliary Evidence and Scope Controls

This final block stores boundary-setting material rather than extra claims: component ablations, intervention audits, prompt and verbalizer sweeps, and large adapter grids. These rows answer scope, localization, or surface-sensitivity questions around the central recovery chain.

Auxiliary group	Scope role
Ablations and patching	Localize candidate mechanisms; full-split sequence-score recovery is evaluated in the sequence-score tables.
Prompt controls	Reported as auxiliary controls for surface sensitivity; per-example gains and nulls evaluate recovery beyond label priors.
Large adapter grids	Map a broad engineering search space; the main text uses selected label-free summaries.
Pythia model-family rows	Serve as model-family controls, including *Pythia-2.8B/12B* controlled-logic rows and external-task rows whose null checks and stability diverge from the main *Qwen3.5* pattern.

Table 47: **Scope-control map.** Auxiliary controls separate mechanism localization, surface sensitivity, engineering sweeps, and model-family controls from the main recovery evidence.

The auxiliary and control rows specify which links are required before a row supports sequence-score recovery: collapsed sequence scores need per-example gains, probes need sequence-score follow-up, and unlabeled offset correction needs shortcut and label-count controls. Read the appendix in the order of Table 7: setup and surface controls first, then the probe/sequence gap, sequence-score correction rows, permutation checks, adapter audits, and scope controls.

ID N	Full Acc.	TF-IDF Acc.	Full ΔAdapt.	TF-IDF ΔAdapt.	Full std	TF-IDF std
25	0.585	0.593	+0.003	+0.028	0.018	0.044
50	0.589	0.603	+0.007	+0.037	0.018	0.029
100	0.582	0.587	-0.000	+0.022	0.017	0.028
250	0.588	0.591	+0.006	+0.025	0.012	0.016
500	0.587	0.590	+0.005	+0.025	0.007	0.010
1000	0.588	0.591	+0.006	+0.026	0.000	0.000

Table 39: **ANLI sequence-score/adapter hybrid sample efficiency.** Supplemental rows fit the ANLI hybrid from unlabeled in-domain scores and evaluate on R2/OOD; small-budget gains make ANLI a distributional stress check.

Radius	Priors	Full Worst	TF-IDF Worst	Full ΔAdapt.	TF-IDF ΔAdapt.	TF-IDF Null Gap
0.025	7	0.571	0.570	-0.011	+0.005	+0.224
0.050	19	0.565	0.558	-0.017	-0.008	+0.221
0.100	61	0.549	0.508	-0.033	-0.058	+0.204

Table 40: **ANLI sequence-score/adapter hybrid prior sensitivity.** Prior perturbations test the ANLI hybrid score distribution; TF-IDF-missed gaps guard against a favorable class prior.

Model	Cal. Acc.	Null Acc. [95%]	Obs.-Null	Cal. Macro-F1	p_{perm}
Pythia-410M	0.333	0.333 [0.305, 0.362]	-0.000	0.333	0.5138
Qwen3.5-0.8B	0.381	0.333 [0.305, 0.362]	+0.048	0.378	0.0017
Pythia-1.4B	0.334	0.333 [0.304, 0.362]	+0.001	0.333	0.4947
Qwen3.5-4B	0.571	0.333 [0.305, 0.362]	+0.238	0.573	0.0001

Table 41: **ANLI sequence-score label-count permutation.** The sequence-score ANLI row is compared against permuted predictions with the same label counts; positive *Qwen3.5* gaps support example-level recovery on adversarial NLI.

Model	Fit N	Eval N	Raw Acc.	Cal. Acc. [95% CI]	Δ [95% CI]	Macro-F1
Qwen3.5-2B	203	1001	0.324	0.472 [0.441, 0.503]	+0.148 [+0.102, +0.194]	0.461
Qwen3.5-4B	203	1001	0.335	0.565 [0.534, 0.597]	+0.231 [+0.186, +0.276]	0.553
Qwen3.5-9B	203	1001	0.348	0.638 [0.608, 0.668]	+0.291 [+0.247, +0.335]	0.632

Table 42: **FOLIO diagnostic label-bias correction.** The fit uses 203 validation examples and evaluates on a separate prepared 1,001-example train surface; this is a diagnostic transfer check, not a standard leaderboard estimate.

Model	Raw Acc.	25 ID	50 ID	100 ID	203 ID	Pos@25
Qwen3.5-2B	0.324	0.454±0.018 (+0.082)	0.453±0.013 (+0.095)	0.458±0.010 (+0.119)	0.472±0.000 (+0.148)	30/30
Qwen3.5-4B	0.335	0.557±0.014 (+0.185)	0.558±0.011 (+0.202)	0.566±0.007 (+0.219)	0.565±0.000 (+0.231)	30/30
Qwen3.5-9B	0.348	0.625±0.026 (+0.204)	0.632±0.015 (+0.256)	0.638±0.009 (+0.266)	0.638±0.000 (+0.291)	30/30

Table 43: **FOLIO diagnostic sample efficiency.** The small fit set makes this a diagnostic stress test rather than a standard benchmark estimate; *Qwen3.5* rows stay positive under smaller unlabeled budgets.

Model	Cal. Acc.	Null Acc. [95%]	Obs.-Null	Cal. Macro-F1	p_{perm}
Qwen3.5-2B	0.472	0.337 [0.308, 0.366]	+0.135	0.461	0.0001
Qwen3.5-4B	0.565	0.340 [0.311, 0.369]	+0.226	0.553	0.0001
Qwen3.5-9B	0.638	0.337 [0.308, 0.366]	+0.301	0.632	0.0001

Table 44: **FOLIO diagnostic label-count permutation.** Label-count permutations on the prepared train surface test whether diagnostic gains exceed label-histogram effects despite the 203-example fit.

Model	Raw Acc.	Worst Acc.	Median Acc.	Worst Δ	Min Macro-F1
Qwen3.5-2B	0.324	0.419	0.457	+0.095	0.421
Qwen3.5-4B	0.335	0.536	0.563	+0.202	0.534
Qwen3.5-9B	0.348	0.607	0.637	+0.260	0.608

Table 45: **FOLIO diagnostic prior sensitivity.** Prior perturbations test whether the 203-example validation fit remains stable under local target-prior shifts.

Model	N	Raw Acc.	Cal. Acc.	Obs.-Null	Macro-F1
Qwen3.5-2B	614	0.316	0.471	+0.137 ($p = 0.0001$)	0.472
Qwen3.5-4B	614	0.324	0.550	+0.218 ($p = 0.0001$)	0.552
Qwen3.5-9B	614	0.337	0.624	+0.282 ($p = 0.0001$)	0.618

Table 46: **FOLIO diagnostic TF-IDF-missed check.** The 203-example validation fit is evaluated on the 1,001-example train surface restricted to TF-IDF-missed examples, testing recovery beyond lexical cues.